

Dr. Fares AL-Suwalih

Exercises 102

$$[1] \lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1}$$

Ans $\rightarrow 3$

$$[2] \lim_{x \rightarrow \infty} \frac{\ln x}{x}$$

Ans $\rightarrow 0$

$$[3] \lim_{x \rightarrow 0} \frac{e^x - 1}{x}$$

Ans $\rightarrow 1$

$$[4] \lim_{x \rightarrow 0} \frac{\sin x}{x}$$

$\rightarrow 1$

$$[5] \lim_{x \rightarrow 0^+} x \ln x$$

$\rightarrow 0$

$$[6] \lim_{x \rightarrow 0^+} \sin^{-1} x \ln x$$

$\rightarrow 0$

$$[7] \lim_{x \rightarrow \infty} (\tanh x)^x$$

$\rightarrow 1$

$$[8] \lim_{x \rightarrow 1^+} \left(\frac{y}{\pi} \tan^{-1}(x) \right)^{\frac{1}{\ln x}}$$

$\rightarrow \frac{2}{\pi} e$

$$[9] \lim_{x \rightarrow \infty} \frac{\ln(7x+9)}{\ln(10x+100)}$$

$\rightarrow 1$

$$[10] \lim_{x \rightarrow \infty} x \sin\left(\frac{1}{x}\right)$$

$\rightarrow 1$

$$[11] \lim_{x \rightarrow \infty} \frac{x + \cos x}{x}$$

$\rightarrow 1$

$$[12] \lim_{x \rightarrow 0^+} \left[\frac{1}{x} - \frac{1}{\sin x} \right]$$

$\rightarrow 0$

$$[13] \lim_{x \rightarrow \infty} (\sqrt{x+1} - \sqrt{x})$$

$\rightarrow 0$

$$[14] \lim_{x \rightarrow \infty} [\ln(2x+1) - \ln(x+7)]$$

$\rightarrow \ln 2$

Show that $\lim_{x \rightarrow 0^+} \left(\frac{5^x + 7^x}{2} \right)^{1/x} = \sqrt{35}$

$$[16] \lim_{x \rightarrow \infty} \left(\frac{x-1}{x+1} \right)^{2x+1} \rightarrow e^{-4}$$

$$[17] \lim_{x \rightarrow \infty} \left(\frac{x+1}{x-1} \right)^x \rightarrow e^2$$

$$[18] \lim_{x \rightarrow 0} \left(\frac{e^x}{\sin x} - \frac{1}{x} \right) \rightarrow 1$$

$$[19] \lim_{x \rightarrow \infty} \left(\frac{x+3}{\cosh x} \right)^{1/x} \rightarrow \frac{1}{e} = e^{-1}$$

$$[20] \lim_{x \rightarrow 0^+} \frac{x \sinh x}{1 - \cosh x} \rightarrow -2$$

[21] For what values of a is the following equation true

$$\lim_{x \rightarrow \infty} \left(\frac{x+a}{x-a} \right)^x = e?$$

Sol: $a = \frac{1}{2}$

$$[22] \lim_{x \rightarrow 0^+} (1+x)^{1/x} \rightarrow e$$

$$[23] \lim_{x \rightarrow 1} \left[\frac{1}{x-1} - \frac{1}{\ln x} \right] \rightarrow \frac{1}{2}$$

$$[24] \lim_{x \rightarrow \infty} \frac{\sin x}{x} \rightarrow 0$$

$$[1] \int \cos(\ln x) dx \xrightarrow{\text{Ans}} = \frac{1}{2} (x \cos(\ln x) + x \sin(\ln x)) + C$$

$$[2] \int e^x \cos x dx \rightarrow = \frac{1}{2} e^x [\sin x + \cos x] + C$$

$$[3] \int x \sec^{-1} x dx \rightarrow = \frac{1}{2} x^2 \sec^{-1} x - \frac{1}{2} (x^2 - 1)^{1/2} + C$$

$$[4] \int \tan^{-1}(\frac{1}{x}) dx \rightarrow = x \tan^{-1}(\frac{1}{x}) + \frac{1}{2} \ln|x^2 + 1| + C$$

$$[5] \int \sin x \ln(\sin x) dx \rightarrow = -\cos x \ln(\sin x) + \ln|\csc x - \cot x| + \cos x + C$$

$$[6] \int (\ln x)^2 dx \rightarrow = x (\ln x)^2 - 2 [x \ln x - x] + C$$

$$[7] \int \frac{\ln(x^2 e^x)}{x^3} dx \rightarrow = -\frac{1}{x^2} \ln x - \frac{1}{2} x^{-2} - x^{-1} + C$$

$$[8] \int x 3^x dx \rightarrow = \frac{x}{\ln 3} 3^x - \frac{1}{(\ln 3)^2} 3^x + C$$

$$[9] \int \sin x \sinh(2x) dx \rightarrow = \frac{1}{5} [-\cos x \sinh(2x) + 2 \sin x \cosh(2x)] + C$$

$$[10] \int e^x \cos(2x) dx \rightarrow = \frac{1}{5} e^x [\cos(2x) + 2 \sin(2x)] + C$$

$$[11] \int e^{\sqrt{x}} dx \rightarrow = 2 (\sqrt{x} e^{\sqrt{x}} - e^{\sqrt{x}}) + C$$

$$[12] \int \sin \sqrt{1+2x} dx \rightarrow = -\sqrt{1+2x} \cos(\sqrt{1+2x}) + \sin(\sqrt{1+2x}) + C$$

$$[13] \int \ln(1+\sqrt{x}) dx \xrightarrow{\text{Ans}} = (\sqrt{x})^2 \ln(1+\sqrt{x}) - \frac{1}{2} (\sqrt{x})^2 + \sqrt{x} - \ln|1-\sqrt{x}| + C$$

$$[14] \int x (\ln x + e^x) dx \xrightarrow{\text{Ans}} = \frac{1}{2} x^2 \ln(x) - \frac{1}{4} x^2 + x e^x - e^x + C$$

$$[15] \int \frac{\cos^{-1}(\ln x)}{x} dx \xrightarrow{\text{Ans}} = \ln x (\cos^{-1}(\ln x) - \sqrt{1 - (\ln x)^2}) + C$$

[16] Suppose f continuously diff. on $[0,1]$ and $f(1) = -3$
 $\int_0^1 x \sqrt{1-f(x)} dx = 5$, find $\int_0^1 \frac{x^2 f'(x)}{\sqrt{1-f(x)}} dx \xrightarrow{\text{Ans}} = 16.$

[17] $\int_0^3 x \sin(x^2) dx \xrightarrow{\text{Ans}} = \frac{1}{2} [-x^2 \cos x^2 + \sin x^2] + C$

- [1] $\int \sin^4 \theta \cos^3 \theta d\theta \rightarrow = \frac{1}{5} (\sin \theta)^5 - \frac{1}{7} (\sin \theta)^7 + C$
 - [2] $\int \frac{\sin^3 2x}{\sec^4 x} dx \rightarrow = -8 \left[\frac{1}{8} (\cos x)^8 - \frac{1}{10} (\cos x)^{10} \right] + C$
 - [3] $\int \sin^3 x \cos\left(\frac{x}{2}\right) dx \rightarrow = -16 \left[\frac{1}{5} u^5 - \frac{1}{7} u^7 \right] + C, u = \cos$
 - [4] $\int (\cos x)^{-1/2} \sin^3 x dx \rightarrow = -2\sqrt{u} + \frac{2}{5} u^{5/2} + C, u = \cos x$
 - [5] $\int \sin^2 x \cos^2 x dx \rightarrow = \frac{1}{8} \left[x - \frac{\sin 4x}{4} \right] + C$
 - [6] $\int \sin^4 x dx \rightarrow = \frac{1}{4} \left[x - \frac{2 \sin 2x}{2} + \frac{1}{2} x + \frac{1}{2} \frac{\sin 4x}{4} \right] + C$
 - [7] $\int_{\pi} \cos^5 x dx \rightarrow = u - \frac{2}{3} u^3 + \frac{1}{5} u^5 + C, u = \sin x$
 - [8] $\int_0 \sin x \cos x \cos(3x) dx \rightarrow = \frac{1}{4} \left[\frac{2}{5} - 2 \right] = -\frac{2}{5}$
 - [9] $\int \sin(3x) \sin(5x) dx \rightarrow = \frac{1}{2} \left[\frac{\sin 2x}{2} - \frac{\sin 8x}{8} \right] + C$
 - [10] $\int \cos(3x) \cos^2(4x) dx \rightarrow = \frac{1}{2} \left[\frac{\sin 3x}{3} \right] + \frac{1}{4} \left[\frac{\sin 11x}{11} + \frac{\sin 5x}{5} \right] + C$
 - [11] $\int \frac{\csc^4 x}{\sqrt[3]{\cot^2 x}} dx \rightarrow = - \left[3 u^{1/3} + \frac{3}{7} u^{7/3} \right] + C, u = \cot x$
 - [12] $\int \operatorname{sech}^6 x dx \rightarrow = u - \frac{2}{3} u^3 + \frac{1}{5} u^5 + C, u = \tanh x$
 - [13] $\int \tanh^3 x dx \rightarrow = \ln |u| + \frac{1}{2} u^{-2} + C, u = \cosh x$
 - [14] $\int \tan^2 x \sec x dx \rightarrow = \sec x \tan x - \frac{1}{2} \ln |\sec x + \tan x| + C$
 - [15] $\int \frac{\tan^3 x \sec^4 x + \sin x}{\sec^3 x} dx \rightarrow = \frac{1}{3} (\sec x)^3 - (\sec x) - \frac{1}{4} (\cos x)^4 + C$
 - [16] $\int (\sin x)^{-1/2} (\cos x)^{-3/2} dx \rightarrow = 2 u^{1/2} + C, u = \tan x$
 - [17] $\int \csc^3 x dx$
 - [18] $\int \cot^2 x \csc x dx$
 - [19] $\int \tan^2 x \sec^3 x dx$
- } by parts

$$\boxed{1} \int \frac{dx}{x^2 \sqrt{4-x^2}} \rightarrow -\frac{1}{4} \cot(\sin^{-1}(\frac{x}{2})) + C$$

$$\boxed{2} \int \frac{x^2}{\sqrt{16-x^2}} dx \rightarrow 8(\sin^{-1}(\frac{x}{4})) - \frac{8x}{4} \frac{\sqrt{16-x^2}}{4} + C$$

$$\boxed{3} \int \frac{\sqrt{x^2-25}}{x} dx \rightarrow 5\sqrt{x^2-25} - 5\sec^{-1}(\frac{x}{5}) + C$$

$$\boxed{4} \int \frac{dx}{(x^2+36)^2} dx \rightarrow \frac{1}{432} \tan^{-1}(\frac{x}{6}) + \frac{\frac{x}{\sqrt{x^2+36}} \cdot \frac{6}{\sqrt{x^2+36}}}{432} + C$$

$$\boxed{5} \int \frac{1}{x^2(x^2+1)^{3/2}} dx \rightarrow -\frac{\sqrt{x^2+1}}{x} - \frac{x}{\sqrt{x^2+1}} + C$$

$$\boxed{6} \int \frac{dx}{x^2-2x+5} \rightarrow \frac{1}{2} \tan^{-1}(\frac{x-1}{2}) + C$$

$$\boxed{7} \int \frac{dx}{\sqrt{2-4x-2x^2}} \rightarrow \frac{1}{\sqrt{2}} \sin^{-1}(\frac{x+1}{\sqrt{2}}) + C$$

$$\boxed{8} \int \frac{e^x}{\sqrt{e^{2x}+2e^x+2}} dx \rightarrow \ln|\sqrt{(e^x+1)^2+1} + (e^x+1)| + C$$

$$\boxed{9} \int \frac{x}{x^2-6x+13} dx \rightarrow \frac{1}{2} \ln|(x-3)^2+4| + \frac{3}{2} \tan^{-1}(\frac{x-3}{2}) + C$$

$$\boxed{10} \int \frac{dx}{\sqrt{3x-x^2}} \rightarrow \sin^{-1}(\frac{x-\frac{3}{2}}{\frac{3}{2}}) + C$$

$$\boxed{11} \int (x^2+4x+3)^{-3/2} dx \rightarrow \frac{-(x+2)}{\sqrt{(x+2)^2-1}} + C$$

$$\boxed{12} \int \frac{x^2+6x+9}{\sqrt{7-6x-x^2}} dx \rightarrow 8\sin^{-1}(\frac{x+3}{4}) - 8\frac{x+3}{4} \cdot \frac{\sqrt{16-(x+3)^2}}{4} + C$$

$$\boxed{13} \int x \sqrt{8-2x-x^2} dx \rightarrow \left[9\left(\frac{\sqrt{9-u^2}}{3}\right)^3 - \frac{9}{2} \sin^{-1}\left(\frac{u}{3}\right) - \frac{18}{4} \frac{u}{3} \frac{\sqrt{9-u^2}}{3} + C \right]$$

$u = x+4$

$$\boxed{1} \int \frac{dx}{x^2-x-6} \xrightarrow{\text{Ans}} \frac{1}{5} \ln|x-3| - \frac{1}{5} \ln|x+2| + C$$

$$\boxed{2} \int \frac{x^2+2x-1}{x^3-x} dx \xrightarrow{\text{Ans}} \ln|x| + \ln|x-1| - \ln|x+1| + C$$

$$\boxed{3} \int \frac{x^2+x-2}{3x^3-x^2+3x-1} dx \xrightarrow{\text{Ans}} \frac{-7}{15} \ln|3x-1| + \frac{2}{5} \ln|x^2+1| + \frac{3}{5} \tan^{-1}(x)$$

$$\boxed{4} \int \frac{3x^4+3x^3-5x^2+x-1}{x^2+x-2} dx \xrightarrow{\text{Ans}} x^3+x + \frac{1}{3} \ln|x-1| + \frac{1}{3} \ln|x+2| + C$$

$$\boxed{5} \int \frac{\cosh x}{\cosh^2 x - 10} dx \xrightarrow{\text{Ans}} \frac{1}{6} \ln|\sinh x - 3| - \frac{1}{6} \ln|\sinh x + 3| + C$$

$$\boxed{6} \int \frac{\sec^4 x}{\sec^2 x - 10} dx \xrightarrow{\text{Ans}} \tan x + \frac{10}{6} \ln|\tan x - 3| - \frac{10}{6} \ln|\tan x + 3| + C$$

$$\boxed{7} \int \frac{2x}{(x^2+1)(x-1)^2} dx \xrightarrow{\text{Ans}} -\frac{1}{(x-1)} - \tan^{-1} x + C$$

$$\boxed{8} \int \frac{x^4-2x^2+4x+1}{x^3-x^2-x+1} dx \xrightarrow{\text{Ans}} \frac{1}{2}x^2+x+\ln|x-1| - 2\frac{1}{(x-1)} - \ln|x+1| + C$$

$$\boxed{9} \int \frac{2x^2-10x+5}{x^3-4x^2+5x} dx \xrightarrow{\text{Ans}} \ln|x| + \frac{1}{2} \ln|x-2| + 4 \tan^{-1}(x-2) + C$$

$$\boxed{10} \int \frac{2-2x}{(x^2+1)(x+1)^2} dx \xrightarrow{\text{Ans}} -\frac{1}{2} \ln|x^2+1| - \tan^{-1} x + \ln|x-1| - 2\frac{1}{(x+1)} + C$$

$$\boxed{11} \int \frac{x^5}{x^4-16} dx \xrightarrow{\text{Ans}} \frac{x^2}{2} + \ln \left| \frac{(x-2)(x+2)}{x^2+4} \right| + C$$

$$\boxed{12} \int \frac{e^{2x}}{e^{2x}+3e^x+2} dx \xrightarrow{\text{Ans}} 2 \ln|u+2| - \ln|u+1| + C, u=e^x$$

$$\boxed{13} \int \frac{\sqrt{x}-6\sqrt{x}}{\sqrt{x}(x+4\sqrt{x})} dx \xrightarrow{\text{Ans}} \frac{3}{2} \ln \left| x^{4/6} + 4 \right| - \frac{3}{2} \tan^{-1} \left(\frac{x^{1/3}}{2} \right) + C$$

$$\boxed{14} \int \frac{dx}{\tan x + \sin x} \xrightarrow{\text{Ans}} \frac{1}{2} \ln \left| \tan \left(\frac{x}{2} \right) \right| - \frac{1}{4} \tan^2 \left(\frac{x}{2} \right) + C$$

Dr. Fares [7.5] Exercises

$$(1) \int \frac{\sin^{\frac{5}{2}} x}{\sec^3 x} dx \xrightarrow{\text{Ans}} \frac{2}{7} \sin^{\frac{7}{2}} x - \frac{2}{11} \sin^{\frac{11}{2}} x + C$$

$$(2) \int \tanh^5 x \operatorname{sech} x dx \xrightarrow{\text{Ans}} -\operatorname{sech} x + \frac{2}{3} \operatorname{sech}^3 x - \frac{1}{5} \operatorname{sech}^5 x + C$$

$$(3) \int \cos(\ln x) dx \xrightarrow{\text{Ans}} \frac{1}{2} (x \cos(\ln x) + x \sin(\ln x)) + C$$

$$(4) \int \frac{x^{\frac{1}{2}} - 1}{x(x^{\frac{1}{6}} + x^{\frac{1}{2}})} dx \xrightarrow{\text{Ans}} 6 \frac{1}{x^{\frac{1}{6}}} + 3 \ln |x^3 + 1| + 6 \tan^{-1}(x^{\frac{1}{6}}) + C$$

$$(5) \int \frac{x dx}{\sqrt{8+2x^2-x^4}} \xrightarrow{\text{Ans}} \frac{1}{2} \sin^{-1}\left(\frac{x^2-1}{3}\right) + C$$

$$(6) \int \frac{dx}{x\sqrt{\sqrt{x}-4}} \xrightarrow{\text{Ans}} 2 \sec^{-1}\left(\frac{\sqrt[4]{x}}{2}\right) + C$$

$$(7) \int \frac{\sec x}{2\sec x + \tan x + 1} dx \xrightarrow{\text{Ans}} \sqrt{2} \tan^{-1}\left(\frac{\tan(\frac{x}{2}) + 1}{\sqrt{2}}\right) + C$$

$$(8) \int \cos(3x) \sin^2 x dx \xrightarrow{\text{Ans}} \left(\frac{1}{6} \sin(3x) - \frac{1}{4} \frac{1}{5} \sin(5x) - \frac{1}{4} \sin x + C \right)$$

$$(9) \int \frac{2 + \tan(\frac{x}{2})}{2\sin x + 2\cos x + 3} dx \rightarrow 2 \ln |\sqrt{(u+2)^2}| - 4 \tan^{-1}(u+2) + 4 \tan^{-1}(u+1) + C$$

$$(10) \int \sqrt{1 - \sin x} dx \xrightarrow{\text{Ans}} 2 \sqrt{1 + \sin x} + C$$

Determine wheather the integral convergent or diverg

$$[1] \int_{-\infty}^0 \frac{1}{(x-8)^{2/3}} dx \xrightarrow{\text{Ans}} \text{Div.}$$

$$[2] \int_{-\infty}^{10} \frac{1}{(x-8)^{2/3}} dx \xrightarrow{\text{Ans}} \text{Div.}$$

$$[3] \int_{-\infty}^{\infty} \cos^2 x dx \xrightarrow{\text{Ans}} \text{Div.}$$

$$[4] \int_{-\infty}^{\infty} \text{sech } x dx \xrightarrow{\text{Ans}} \text{Conv.} = \pi$$

$$[5] \int_4^{\infty} \frac{x+18}{x^2+x-12} dx \xrightarrow{\text{Ans}} \text{Div.}$$

$$[6] \int_0^1 \frac{e^{\sqrt{x}}}{\sqrt{x}} dx \xrightarrow{\text{Ans}} \text{Conv.} = 2e-2$$

$$[7] \int_0^1 x \ln x dx \xrightarrow{\text{Ans}} \text{Conv.} = -\frac{1}{4}$$

$$[8] \int_{1/e}^e \frac{1}{x (\ln x)^2} dx \xrightarrow{\text{Ans}} \text{Div.}$$

$$[9] \int_0^9 \frac{x}{3\sqrt{x-1}} dx \xrightarrow{\text{Ans}} \text{Conv.} = \frac{243}{10}$$

$$[10] \int_0^{\pi} \frac{\sec^2 x}{4 + \tan^2(x)} dx \xrightarrow{\text{Ans}} \text{Conv.} = \frac{\pi}{2}$$

$$[11] \int_0^{\infty} x e^{-x} dx \xrightarrow{\text{Ans}} \text{Conv.} = 1$$

1] let $C: y = \frac{x^3}{3} + \frac{1}{4x}$, $x \in [1, 2]$

(a) Find the length of curve? $\rightarrow$ Ans: $\boxed{\frac{59}{24}}$

(b) Find the area of surface obtained by rotating C about y -axis. $\rightarrow$ Ans: $\boxed{\frac{\pi}{2} [15 + \ln 2]}$

2] let $C: x = \frac{1}{3} (y^2 + 2)^{3/2}$, $y \in [1, 2]$

(a) Find the length of curve? $\rightarrow$ Ans: $\boxed{\frac{10}{3}}$

(b) Find area of surface obtained by rotating C about x -axis? $\rightarrow$ Ans: $\boxed{\frac{21\pi}{2}}$

3] let $C: y = \cosh x$, $0 \leq x \leq 1$

(a) Find the length of curve? $\rightarrow$ Ans: $\boxed{\frac{e - e^{-1}}{2}}$

(b) Find the area of C is rotated
 about x -axis: $\text{Ans: } \frac{\pi}{2} [e^2 + 2 - \frac{1}{2e}]$
 about y -axis: $\text{Ans: } [2\pi(1 - \frac{1}{e})]$

4] Find the length of the curve:

$y = \ln\left(\frac{e^x + 1}{e^x - 1}\right)$, $a \leq x \leq b$, $a > 0$

Ans: $\ln |\sinh(b)| - \ln |\sinh(a)|$

5] let $y = \sqrt{x+1}$, $1 \leq x \leq 5$

Find area about x -axis? $\text{Ans: } \left(\frac{\pi}{6} (125 - 27) \right)$

6] Find area about y -axis:

$y = \frac{x^2}{8} - \ln x$, $x \in [1, 2]$ (Ans: $\frac{19\pi}{6}$)

7] let $C: y = \ln(\sec x)$, $x \in [0, \frac{\pi}{4}]$

Find the length of curve? (Ans: $\ln(\sqrt{2} + 1)$)

8] Find the arc length of the graph of $f(x)$

$f(x) = 2 \int_1^{\sqrt{x}} t \sqrt{t^4 - 1} dt$, from $x=1$ to $x=4$?

Ans: $\sqrt{15}$

[1] Find Centroid of region bounded by

$$y = \sqrt{x-1}, \quad x\text{-axis}, \quad x=5?$$

Ans: $(\bar{x}, \bar{y}) = (\frac{17}{5}, \frac{3}{4})$

[2] Find Centroid of $y = \sqrt{1-x^2}, \quad y=0$

Ans: $(\bar{x}, \bar{y}) = (0, \frac{4}{3\pi})$

^{mo} [3] Find Centroid of $y = \sin x, \quad y = \cos x, \quad x=0, \quad x=\frac{\pi}{4}$

Ans: $(\bar{x}, \bar{y}) = (\frac{1}{\sqrt{2}-1} (\frac{\pi}{4}(\sqrt{2})-1), \frac{1}{4(\sqrt{2}-1)})$

[4] Find Centroid $y = 2x^2, \quad y = x^3$

Ans: $(\bar{x}, \bar{y}) = (\frac{6}{5}, \frac{(47)(8)}{35})$

[5] Find Centroid $x = y^2, \quad x=4$

Ans: $(\bar{x}, \bar{y}) = (\frac{12}{5}, 0)$

[6] Find Centroid $x = y^2, \quad y = x^2$

Ans: $(\bar{x}, \bar{y}) = (\frac{9}{20}, \frac{1}{4})$

ii] Find Center of mass

$$y = \sec^2 x, \quad x\text{-axis}, \quad y\text{-axis}, \quad \text{and} \quad x = \frac{\pi}{4}$$

Ans: $(\bar{x}, \bar{y}) = (\frac{\pi - 2\ln 2}{4}, \frac{2}{3})$

[7] $x = 5 - y^2$ and $x=0$?

Ans: $(\bar{x}, \bar{y}) = (2, 0)$

[8] $y = \frac{1}{x}, \quad y=0, \quad x=1, \quad x=2$?

Ans: ~~$(\bar{x}, \bar{y}) = (\frac{1}{\ln 2}, \frac{1}{\ln 2})$~~
 $(\bar{x}, \bar{y}) = (1, \frac{1}{\ln 2})$

[*] let $C: x = g(t), y = h(t), a \leq t \leq b$

[1] $y' = \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \rightarrow \frac{\text{نسبة}}{\text{المقام}}$

[2] Horizontal tangent line (H.T.L.) $\text{نسبة} = 0$

$$\frac{dy}{dt} = 0 \quad \text{and} \quad \frac{dx}{dt} \neq 0$$

[3] vertical tangent line (V.T.L.) $\text{المقام} = 0$

$$\frac{dx}{dt} = 0, \quad \frac{dy}{dt} \neq 0$$

[4] The length of curve

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

[5] Area about x-axis:

$$S = \int_a^b 2\pi y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$y(t)$

[6] Area about y-axis:

$$S = \int_a^b 2\pi x \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$x(t)$

[7] $y'' = \frac{d^2y}{dx^2} = \frac{d^2y/dt^2}{dx/dt} = \frac{\frac{d}{dt} \left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$

نسبة / المقام

هذه قوائمنا التي يمكن استخدامها في بعض الأحيان لإيجاد مساحة دالة وانحنى
ونقول ان Arc length طول المنحنى فليكن y دالة في x فواضنا y
التي $9.1, 9.2$ راجع في صعوبة إيجاد y فليكن x و y
عن بعضهما الآخر لذلك نحتاج إلى Parameters
Arc length

[1] let $C: X = e^t \cos t, y = e^t \sin t, t \in [0, 2\pi]$

(a) Find eq of T.L. at $t = \frac{\pi}{2} \xrightarrow{\text{Ans}} y - e^{\frac{\pi}{2}} = -x$

(b) Find pts at which T.L. is (i) Vertical

Ans: (i) $(\frac{e^{\pi/4}}{\sqrt{2}}, \frac{e^{\pi/4}}{\sqrt{2}}), (-\frac{e^{5\pi/4}}{\sqrt{2}}, \frac{e^{5\pi/4}}{\sqrt{2}})$ (ii) Horizontal

Ans: (ii) $(\frac{e^{-3\pi/4}}{\sqrt{2}}, \frac{e^{-3\pi/4}}{\sqrt{2}}), (\frac{e^{3\pi/4}}{\sqrt{2}}, -\frac{e^{3\pi/4}}{\sqrt{2}})$

(c) Find the length of curve?

Ans: $\sqrt{2} (e^{2\pi} - 1)$

(d) Find area of surface obtained by rotating C about y -axis? Ans: $\frac{2\pi\sqrt{2}}{5} [2e^{4\pi} - 2]$

[2] let $C: X = t - \sin t, y = 1 - \cos t, 0 \leq t \leq 2\pi$

(a) Determine if C is concave upward or downward at $t = \pi$? Ans: concave downward $\cap$

(b) Find area of surface generated by rotating C about x -axis? Ans: $\frac{64\pi}{3}$

[3] let $C: X = 2e^t - 3t, y = e^t + 5t$

Find pts on curve C at which the T.L. is Parallel to the line $y = 7x + 1$? Ans: $(x, y) = (4 - \ln 8, 2 + \ln 32)$

[4] let $C: X = e^t - t, y = 2e^t, t \in \mathbb{R}$

(a) Find pts (if any) at which the T.L. is

(i) Vertical $\rightarrow$ Ans: (1, 2)

(ii) horizontal $\rightarrow$ Ans: No solution

(b) Find an interval at which C is $\cup$ or $\cap$ and find inflection pt (if any)? Ans: C is $\cup$ on $(-\infty, 0)$
 C is $\cap$ on $(0, \infty)$

[5] let $C: X = \cos^3 t, y = \sin^3 t, t \in [0, \frac{\pi}{2}]$

Find area of surface generated by rotating C about x -axis? Ans: $\frac{6\pi}{5}$

1] sketch the graph of $r = \sqrt{4 + 5 \cos^2 \theta}$
 $0 \leq \theta \leq 2\pi$ by labeling at least 4 points on it
 and find the area of the region bounded by the
 graph?

2] sketch the graph of $r = \frac{2}{\pi} \theta + 1$, $0 \leq \theta \leq 2\pi$
 by labeling at least 3 points on it and find
 the area of the region bounded by the graph
 in the second quadrant?

3] Given $r = e^{3\theta}$
 a) sketch the graph of the Polar equation?

b) Find the slope of the T. L to the above
 graph at the point that corresponds $\theta = \frac{\pi}{2}$

me 4] Find the length of the curve $r = 4 \cos^2\left(\frac{\theta}{2}\right)$
Ans = 16

me 5] Find the area of the region inside both
 $r = \sin \theta$ and $r = \sin(2\theta)$
Ans
 $A = \frac{\pi}{4} - \frac{3\pi}{16}$

6] Find the area enclosed by one loop of
 the curve $r = \sin(3\theta)$
Ans
 $\frac{\pi}{12}$

7] let $r = 2 + 2 \cos \theta$

a] Find slope of T. L at $\theta = \frac{\pi}{4}$

b] Find eq of T. L at $\theta = \frac{\pi}{4}$ (in rectangular coordinate).

8] Set up an integral for the area of the region
 inside both curves : $r = \sin 2\theta$, $r = \sin \theta$

9] Find the length of the cardioid $r = 1 + \sin \theta$
Ans
 $\sqrt{1-0} = 1$

Exercises 11.1

Dr. Fares

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Find the limits of the following seq. and determine the Convergence

- ① $a_n = \sqrt[n]{3n^3}$ (Ans: 1 Conv)
- ② $a_n = (n+4)^{\frac{1}{n+4}}$ (Ans: 1 Conv)
- ③ $a_n = \left(\frac{1}{3}\right)^n + \frac{1}{\sqrt{2}^n}$ (Ans: 0 Conv)
- ④ $a_n = \frac{3^n 6^n}{2^n n!}$ (Ans: 0 Conv)
- ⑤ $a_n = \frac{n!}{(10)^{6n}}$ (Ans: ∞ Div)
- ⑥ $a_n = \left(\frac{3n+1}{3n-1}\right)^n$ (Ans: $e^{2/3}$ Conv)
- ⑦ $a_n = \left(1 - \frac{1}{n^2}\right)^n$ (Ans: 1 Conv)
- ⑧ $a_n = \sqrt[n]{n^2+n}$ (Ans: 1 Conv)
- ⑨ $a_n = \left(\frac{2n+1}{3n+2}\right)^n$ (Ans: 0 Conv)
- ⑩ $a_n = \left(\frac{-3}{5}\right)^n$ (Ans: 0 Conv)
- ⑪ $a_n = \left(\frac{e}{\pi}\right)^n$ (Ans: 0 Conv)
- ⑫ $a_n = \left(\frac{-3}{2}\right)^n$ (Ans: DNE Div)
- ⑬ $a_n = (-1)^n$ (Ans: DNE Div)
- ⑭ $\left\{ n^2 \sin\left(\frac{2}{n^2}\right) + \frac{\cos^2 n}{3^n} \right\}$ (Ans: 3 Conv)
- ⑮ $\left\{ \frac{n^2 \sin\left(\frac{2}{n}\right)}{\sin n} \right\}$ (Ans: ∞ Div)
- ⑯ $\left\{ \frac{n^3 + 5n}{5n^3 + \cos n + n^{-2}} \right\}$ (Ans: $\frac{1}{5}$ Conv)
- ⑰ $\left\{ \frac{n + \sin n}{\sqrt{4n^2 + 1}} \right\}$ (Ans: $\frac{1}{2}$ Conv)
- ⑱ $\left\{ \frac{(2n-1)!}{(2n+3)!} \right\}$ (Ans: 0 Conv)
- ⑲ $\left\{ (n-2)! \sin\left(\frac{1}{n!}\right) \right\}$ (Ans: 0 Conv)
- ⑳ $\left\{ (2^n + 3^n)^{1/n} \right\}$ (Ans: 3 Conv)
- ㉑ $\left\{ \left(1 + \frac{3}{n}\right)^{2n^2} \right\}$ (Ans: e^6 Conv)
- ㉒ $\left\{ \left(1 - \frac{\ln 2}{n}\right)^{n^2} \right\}$ (Ans: $\frac{1}{2}$ Conv)
- ㉓ $\left\{ \left(\frac{n}{n+3}\right)^{5n^2} \right\}$ (Ans: e^{-15} Conv)
- ㉔ $\left\{ \left(\frac{n+1}{n+3}\right)^n \right\}$ (Ans: e^{-2} Conv)
- ㉕ $\left\{ (-1)^n \frac{\ln n}{n} \right\}$ (Ans: 0 Conv)
- ㉖ $\left\{ (-1)^n \frac{2n^2 + 1}{n^2 + 5} \tan^{-1} n \right\}$ (Ans: DNE Div)
- ㉗ $\left\{ \frac{3^{n+1} + n^2}{(-3)^n + 2^n} \right\}$ (Ans: DNE Div)
- ㉘ $\left\{ \frac{n^2}{2n+1} \sin\left(\frac{\pi}{n}\right) \right\}$ if n is even (Ans: DNE Div)
if n is odd (Ans: Div)
- ㉙ $\left\{ n^2 \ln\left(1 + \frac{2}{n^2}\right) \right\}$ (Ans: 0 Conv)
- ㉚ $\left\{ \frac{(-1)^n + \sin n}{n+1} \right\}$ (Ans: 0 Conv)
- ㉛ $\left\{ \sqrt[n]{2} + (-1)^n n^2 e^{-n} + n \tan \frac{\pi}{n} \right\}$ (Ans: $\frac{1}{2} + \pi$ Conv)

Exercises 11.2

Dr. Fares

Determine if series is convergent or Divergent. Find

Sum if series is convergent.

① $\sum_{n=2}^{\infty} \frac{2}{4n^2 - 8n + 3}$

(Ans: Conv, $S = 1$)

② $\sum_{n=1}^{\infty} \frac{2}{4n^2 - 1}$

(Ans: Conv, $S = 1$)

③ $\sum_{n=1}^{\infty} \ln(1 + \frac{1}{n})$

(Ans: Div, No sum)

④ $\sum_{n=0}^{\infty} \tan^{-1}(n+2) - \tan^{-1}(n+1)$

(Ans: Conv, $S = \frac{\pi}{4}$)

⑤ $\sum_{n=2}^{\infty} \left(\frac{2}{n^2 - 1} \right)$

(Ans: Conv, $S = \frac{3}{2}$)

⑥ $\sum_{n=2}^{\infty} \frac{1}{2^{\frac{1}{n-1}}} - 2^{\frac{1}{n+1}}$

(Ans: Conv, $S = \sqrt{2}$)

⑦ Find Sum if the series is convergent?

⑦ $\sum_{n=1}^{\infty} (2)^{n-2} (3)^{1-n}$

(Ans: Conv, $S = \frac{3}{2}$)

⑧ $\sum_{n=1}^{\infty} \left(\frac{\pi}{\pi+1} \right)^n$

(Ans: Conv, $S = \pi$)

⑨ $\sum_{n=0}^{\infty} \left(\frac{e}{e+1} \right)^n$

(Ans: Conv, $S = \frac{e+1}{e}$)

⑩ $\sum_{n=1}^{\infty} \left(\frac{e^{n+1}}{2^n} \right)$

(Ans: Div, No sum)

⑪ $\sum_{n=1}^{\infty} \frac{3^n - 2^n}{(-4)^n}$

(Ans: Conv, $S = \frac{-2}{21}$)

⑫ Find value of x for which series is convergent and sums?

⑫ $\sum_{n=1}^{\infty} \frac{(x-3)^n}{2^{n+1}}$

(Ans: Conv, $S = \frac{\frac{1}{2}(\frac{x-3}{2})}{1 - (\frac{x-3}{2})}$
[$x \in (1, 5)$])

⑬ $\sum_{n=1}^{\infty} \frac{2^n}{x^n}$

(Ans: Conv, $S = \frac{2/x}{1 - 2/x}$
[$x \in (2, \infty) \cup (-\infty, -2]$])

⑭ $e^{-x} + e^{-2x} + e^{-3x} + \dots$

(Ans: Conv if $x \in (0, \infty)$, $S = \frac{-x}{1 - e^{-x}}$)

⑮ If $\sum_{n=1}^{\infty} e^{nx} = 10$ find x (Ans: $x = \ln(\frac{10}{11}) = \ln(10) - \ln(11)$)

⑯ let $x = 0.794$
⑰ express x as a sum of a geom. s.
⑱ find integers m and n s.t: $x = \frac{m}{n}$

Ans: $x = \frac{m}{n} = \frac{794}{99}$

Determine if series is Convergent or Divergent.

$$1 \quad \sum_{n=2}^{\infty} \frac{1}{n \ln n}$$

Ans $\rightarrow$ Div.

$$2 \quad \sum_{n=2}^{\infty} \frac{1}{\ln(n)^n}$$

Ans $\rightarrow$ Div

$$3 \quad \sum_{n=1}^{\infty} \frac{n}{e^{n^2}}$$

Ans $\rightarrow$ Conv.

$$4 \quad \sum_{n=1}^{\infty} \frac{1}{n^2} e^{\frac{1}{n}}$$

Ans $\rightarrow$ Conv.

$$5 \quad \sum_{n=1}^{\infty} \frac{e^{-\sqrt{n}}}{\sqrt{n}}$$

Ans $\rightarrow$ Conv.

$$6 \quad \sum_{n=2}^{\infty} \frac{\ln n}{n^2}$$

Ans $\rightarrow$ Conv.

$$7 \quad \sum_{n=1}^{\infty} \frac{1}{n(4 + \ln^2 n)}$$

Ans $\rightarrow$ Conv.

$$8 \quad \sum_{n=2}^{\infty} \ln \sqrt[n]{n}$$

Ans $\rightarrow$ Div.

$$9 \quad \sum_{n=1}^{\infty} \frac{\ln n}{n}$$

Ans $\rightarrow$ Div

Explain why the integral test can't be used to determine whether the series is convergent $\sum_{n=1}^{\infty} \frac{\cos \pi n}{\sqrt{n}}$

Find the values of p for which the series is convergent $\sum_{n=1}^{\infty} n(1+n^2)^p$

Ans $\rightarrow p < -1$

Determine if series is Conv. or Div.

① $\sum_{n=1}^{\infty} \frac{\cos^2 n}{3^n}$	Conv. (C.T)	⑩ $\sum_{n=1}^{\infty} \frac{2n^2+n+1}{3n^3+2n+7}$	Div
② $\sum_{n=1}^{\infty} \frac{\cos n}{n^2}$	Conv. (C.T)	⑪ $\sum_{n=1}^{\infty} \frac{n+\sqrt{n}}{\sqrt[3]{n^7+n^2}}$	Conv.
③ $\sum_{n=1}^{\infty} \frac{2-\cos n}{\sqrt{n-1}}$	Div	⑫ $\sum_{n=1}^{\infty} \frac{2^n}{(10)^n - 2n^{10}}$	Conv.
④ $\sum_{n=1}^{\infty} \frac{\sin \sqrt{n}}{\sqrt[3]{n^4+1}}$	Conv.	⑬ $\sum_{n=1}^{\infty} \frac{3n^2+5n}{e^n(n^2-1)}$	Conv.
⑤ $\sum_{n=3}^{\infty} \frac{1+\sin n}{\sqrt{n^2-4}}$	Div	⑭ $\sum_{n=1}^{\infty} \frac{n+3}{(n+1)\sqrt{n+2}}$	Div
⑥ $\sum_{n=2}^{\infty} \frac{2+\cos n}{\sqrt[3]{n^2-1}}$	Div	⑮ $\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$	Div
⑦ $\sum_{n=1}^{\infty} \frac{\sin n + e^n}{3^n + n^2}$	Conv	⑯ $\sum_{n=1}^{\infty} \cos\left(\frac{1}{n}\right)$	Div
⑧ $\sum_{n=2}^{\infty} \frac{\ln n}{n}$	Div	⑰ $\sum_{n=1}^{\infty} \sin\left(\frac{\pi}{n^2}\right)$	Conv
⑨ $\sum_{n=2}^{\infty} \frac{\ln n}{\sqrt{n}}$	Div	⑱ $\sum_{n=1}^{\infty} \ln\left(1+\frac{1}{3^n}\right)$	Conv.
⑩ $\sum_{n=2}^{\infty} \frac{\ln n}{n^3+n+1}$	Conv	⑲ $\sum_{n=1}^{\infty} \frac{3^n+3^n}{n^2+4^n}$	Conv.
⑪ $\sum_{n=2}^{\infty} \frac{1}{\ln n}$	Div	⑳ $\sum_{n=2}^{\infty} \ln\left(1+\frac{1}{\sqrt{n}}\right)$	Div
⑫ $\sum_{n=2}^{\infty} \frac{(\ln n)^3}{n}$	Div	㉑ $\sum_{n=1}^{\infty} \tan\left(\frac{1}{n}\right)$	Div
⑬ $\sum_{n=1}^{\infty} \frac{1}{e^n - n}$	Conv.	㉒ $\sum_{n=1}^{\infty} \frac{n \tan^{-1}(n) + 1}{n^3 - n + 1}$	Conv.
⑭ $\sum_{n=1}^{\infty} \frac{1+2^n}{3^n+n^2}$	Conv.	㉓ $\sum_{n=1}^{\infty} \frac{n+\ln n}{n^2+1}$	Div

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Exercises 11.6

Test the convergence of the following series:

① $\sum_{n=1}^{\infty} \frac{n! \ln n}{n(n+2)!} \rightarrow \text{Conv.}$

② $\sum_{n=1}^{\infty} \frac{(n!)^n}{(n)^{n^2}} \rightarrow \text{Conv.}$

③ $\sum_{n=1}^{\infty} \frac{4^n (n!)^2}{(2n)!} \rightarrow \text{Div}$

④ $a_1 = 3, a_{n+1} = \frac{n}{n+1} a_n \rightarrow \text{Div}$

⑤ $\sum_{n=1}^{\infty} \frac{n^2}{3^n} \rightarrow \text{Conv.}$

⑥ $\sum_{n=1}^{\infty} \left(\frac{2}{n}\right)^n n! \rightarrow \text{Conv.}$

⑦ $\sum_{n=1}^{\infty} \frac{2^n n^{2n}}{(1+3n^2)^n} \rightarrow \text{Conv.}$

⑧ $\sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^{n^2} \rightarrow \text{Conv.}$

⑨ $\sum_{n=1}^{\infty} 2^n \left(1 + \frac{2}{n}\right)^{n^2} \rightarrow \text{Div}$

⑩ $\sum_{n=1}^{\infty} \frac{(3n)!}{e^{2n}} \rightarrow \text{Div}$

⑪ $\sum_{n=1}^{\infty} (1 + \cos^2 n) \frac{2^n (n)}{(2n+1)^n} \rightarrow \text{Div}$

Dr. Fares Exercises 11.5

Determine whether the series is A.C, C.C, or Div

$$① \sum_{n=1}^{\infty} (-1)^n (\sqrt{n+\sqrt{n}} - \sqrt{n}) \rightarrow \text{Div.}$$

$$② \sum_{n=1}^{\infty} \frac{(-1)^n (2n)!}{2^n n! \cdot n} \rightarrow \text{Div}$$

$$③ \sum_{n=1}^{\infty} (-1)^n \frac{1}{\sqrt[3]{n^2+2}} \rightarrow \text{C.C}$$

$$④ \sum_{n=1}^{\infty} (-1)^n \frac{n}{n^2+1} \rightarrow \text{C.C}$$

$$⑤ \sum_{n=1}^{\infty} \frac{\cos(n\pi)}{n^2+2} \rightarrow \text{A.C}$$

$$⑥ \sum_{n=1}^{\infty} (-1)^n \frac{3n^2+5}{n^{3/2}+n^2+7} \rightarrow \text{A.C}$$

$$⑦ \sum_{n=1}^{\infty} (-1)^n \frac{n! \cdot 2^n}{(2n)!} \rightarrow \text{A.C}$$

$$⑧ \sum_{n=1}^{\infty} (-1)^n \left(\frac{n+3}{n+1} \right)^{n^2} \rightarrow \text{Div}$$

$$⑨ \sum_{n=1}^{\infty} (-1)^n \frac{n + \ln n}{3^n} \rightarrow \text{A.C}$$

$$⑩ \sum_{n=1}^{\infty} (-1)^n \frac{n^2}{5^n} \rightarrow \text{A.C}$$

$$⑪ \sum_{n=1}^{\infty} \frac{\sin n}{4^n} \rightarrow \text{A.C}$$

$$⑫ \sum_{n=1}^{\infty} \frac{2^n \cos(3n)}{(1+\frac{1}{n})^{n^2}} \rightarrow \text{A.C}$$

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EX (11.7)

- [1] Determine whether the seq $a_n = n + \frac{1}{n}$ is $\uparrow$, $\downarrow$, or not monotonic. Is this seq. bounded?

Ans: a_n is $\uparrow$, $\{a_n\}$ Div and unbounded.

- [2] Find $\lim_{n \rightarrow \infty} \sqrt[n]{2^n + 3^n}$ if it exists. Ans: [3]

- [3] Find all values of $\theta \in [0, \pi]$ for which the series $\sum_{n=1}^{\infty} \cos^n \theta$ is conv., and find its sum

Ans: $\theta \in (0, \pi)$, $S = \frac{\cos \theta}{1 - \cos \theta}$

- [4] Determine whether the series $\sum_{n=1}^{\infty} \frac{4}{n^2 + 2n}$ is

conv. or Div., by expressing its partial sum S_n as a telescoping sum. If it's conv., find its sum.

Ans: conv. sum = 3

- [5] Test each series for conv:

(a) $\sum_{n=2}^{\infty} \frac{1}{n \ln^2 n}$ conv

(b) $\sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{1}{\sqrt{n}}$ conv.

- [6] Determine whether it is AC, CC, or Div:

(a) $\sum_{n=1}^{\infty} \frac{\cos(n\pi)}{n^{3/4}}$ C.C

(b) $\sum_{n=1}^{\infty} \left(\frac{n+2}{n} \right)^{n^2}$ Div

[7] True or False:

(a) The seq $1, \frac{1}{2}, 1, \frac{1}{4}, 1, \frac{1}{8}, 1, \frac{1}{16}, \dots$ Conv. to ①

(F)

(b) A conv. seq $\{a_n\}$ that satisfies $a_{n+1} = \sqrt{14a_n}, n \geq 1$

(T) has the limit $L = \frac{1+\sqrt{5}}{2}$

(c) The series $\sum_{n=1}^{\infty} \frac{3^n}{p^n}$ Conv. , $\forall p$ s.t: $|p| \geq 3$

(F)

(d) If $\sum_{n=0}^{\infty} a_n = A$, and $\sum_{n=0}^{\infty} b_n = B$ ($a_n > 0, b_n > 0$), then

(F) $\sum_{n=0}^{\infty} a_n b_n = AB$

[8] Determine A.C, C.C, Div

(a) $\sum_{n=1}^{\infty} \frac{n^2 \sin(2n)}{1+3^n}$ (A.C)

(b) $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n+1}}$ (C.C)

[9] Determine whether the following series is Conv. or Div.

(a) $\sum_{n=1}^{\infty} \frac{\ln n}{n^{3/2}}$ conv

(b) $\sum_{n=1}^{\infty} n \tan \frac{1}{\sqrt{n}}$ Div

(c) $\sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{n}{n+1} \right)$ ~~Div~~ conv

(d) $\sum_{n=2}^{\infty} \left(\frac{n+1}{3n-1} \right)^{3n}$ Conv

[10] Find a power series representation for the function
 $f(x) = \frac{x^3}{(x-2)^2}$ and Determine its radius of conv

Ex: Find the radius and interval of Converge
of Power Series?

$$\boxed{1} \quad \sum_{n=1}^{\infty} \frac{2^n}{n^2} (x+2)^n \xrightarrow{\text{Ans}} I = \left[-\frac{5}{2}, -\frac{3}{2}\right] \\ R = \frac{1}{2}$$

$$\boxed{2} \quad \sum_{n=2}^{\infty} \frac{x^n}{(\ln n)^n} \xrightarrow{\text{Ans}} I = \mathbb{R}, R = \infty$$

$$\boxed{3} \quad \sum_{n=1}^{\infty} n \cdot 4^n (x-1)^{2n} \xrightarrow{\text{Ans}} I = \left(\frac{1}{2}, \frac{3}{2}\right), R = \frac{1}{2}$$

$$\boxed{4} \quad \sum_{n=2}^{\infty} \frac{(-1)^n}{(n \ln n) 3^{2n}} (2x+3)^n \xrightarrow{\text{Ans}} I = [-6, 3], R = \frac{9}{2}$$

$$\boxed{5} \quad \sum_{n=1}^{\infty} \left(\frac{n}{2}\right)^n (x-3)^n \xrightarrow{\text{Ans}} R = 0, x = 3$$

$$\boxed{6} \quad \sum_{n=1}^{\infty} \frac{n! x^n}{(2n)!} \xrightarrow{\text{Ans}} I = \mathbb{R}, R = \infty$$

$$\boxed{7} \quad \sum_{n=0}^{\infty} \frac{n x^n}{4^n (n^2+1)} \xrightarrow{\text{Ans}} I = [-4, 4), R = 4$$

$$\boxed{8} \quad \sum_{n=1}^{\infty} \frac{(-1)^n (2x-3)^n}{2n-1} \xrightarrow{\text{Ans}} I = (1, 2], R = \frac{1}{2}$$

$$\boxed{9} \quad \sum_{n=1}^{\infty} \frac{(x-3)^n}{5^n \sqrt{2n+1}} \xrightarrow{\text{Ans}} I = [-2, 8) / R = 5$$

Dr. Fares Exercises 11.9

Ex: Find Power series representation of $f(x)$ in power of x ?

[1] $f(x) = \ln(3-x)$ $\xrightarrow{\text{Ans}}$ $f(x) = \ln 3 - \sum_{n=1}^{\infty} \frac{x^n}{3^n n}$ $f(x) = \ln 3 - \sum_{n=0}^{\infty} \frac{x^{n+1}}{3^{n+1} (n+1)}$

[2] $f(x) = x^2 \tan^{-1}(2x)$ $\xrightarrow{\text{Ans}}$ $f(x) = \sum_{n=0}^{\infty} (-1)^n \frac{(2)^{2n+1} x^{2n+3}}{(2n+1)}$

[3] $f(x) = \int_0^x e^{-t^2} dt$ $\xrightarrow{\text{Ans}}$ $f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \frac{x^{2n+1}}{(2n+1)}$

[4] $g(t) = \begin{cases} \frac{e^t - 1}{t} & , t \neq 0 \\ 1 & , t = 0 \end{cases}$

$f(x) = \int_0^x g(t) dt$, find power series of $f(x)$?

Ans: $\sum_{n=1}^{\infty} \frac{1}{n!} \frac{x^n}{n}$

[5] $f(x) = \frac{1}{(1-x)^3}$ $\xrightarrow{\text{Ans}}$ $f(x) = \frac{1}{2} \sum_{n=2}^{\infty} n(n-1) x^{n-2}$

[6] $f(x) = \frac{x^2}{(x-2)^2}$ $\xrightarrow{\text{Ans}}$ $f(x) = \sum_{n=1}^{\infty} \frac{n x^{n+1}}{2^{n+1}}$

[7] $f(x) = \frac{4}{3+2x-x^2}$ find power series of $f(x)$ in power of $x-1$?

Ans: $\sum_{n=0}^{\infty} \frac{(x-1)^n}{4^n}$, $(x-1)^2 < 4$

[8] $f(x) = \frac{x}{4^x}$ $\xrightarrow{\text{Ans}}$ $\sum_{n=0}^{\infty} \frac{(-1)^n (\ln 4)^n x^{n+1}}{n!}$

[9] Find sum $\sum_{n=1}^{\infty} (-1)^n \frac{1}{(n+1) 2^{n+1}}$?

[10] (a) $f(x) = \ln(1+x)$ $\xrightarrow{\text{Ans}}$ $\sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1}$ (b) Use (a) to find the sum $\sum_{n=1}^{\infty} \frac{n+2}{n \cdot 2^n} \xrightarrow{\text{Ans}} \ln 2 + \frac{1}{2}$

- ① Find the Maclaurin series of the function:

$$f(x) = \int_0^x \sin(t^2) dt, \text{ and state its radius of}$$

Convergence $\xrightarrow{\text{Ans}}$ $\sum_{n=0}^{\infty} (-1)^n \frac{x^{4n+3}}{(2n+1)!(4n+3)}$

$$R = \infty$$

- ② Find the sum of the series $\sum_{n=1}^{\infty} (-1)^n \frac{\pi^{2n+1}}{4^{2n+1}(2n+1)!}$

(Hint: Use the Maclaurin series for $\sin x$) $\xrightarrow{\text{Ans}} \frac{\sqrt{2}}{2} - \frac{\pi}{4}$

- ③ Find the Taylor poly. of degree 3 of the function $f(x) = \sqrt{1+x}$, centered at $a=0$ $\xrightarrow{\text{Ans}} 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16}$

- ④ Find the Fourth Maclaurin poly for $f(x) = \sin(2x)$

$$\text{Ans: } 2x - \frac{4}{3}x^3$$

- ⑤ Find the Taylor series for $f(x) = \ln x$ at $a=2$

$$\text{Ans: } \ln(2) + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(x-2)^n}{2^n \cdot n}$$

- ⑥ Find the Maclaurin series for $f(x) = e^{5x}$ at $a=$

$$\text{Ans: } \sum_{n=0}^{\infty} 5^n \frac{x^n}{n!}$$